Problem Set 5

1

a. The total number of games for tic tac toe game is 9! which is 362880. However, we need to stop playing game when the game is over. Thus, we need to deduct some numbers which is placed after the game is over or we need to find the number of games that is over in 5 moves, 6 moves,7, 8, and 9 moves.

Let x starts the game and there are 3 verticals, 3 horizontals and 2 diagonals = total 8 lines

For the number of games that is over in 5 moves, there will be 2 'X', which is factorial of 3 and 2 "O", which goes into remaining 6 squares (9- 3 = 6, so it's 6 * 5 = 30) on the game. Thus, the possibilities for the number of games that is over in 5 moves is 8 * 3 * 2 * 6 * 5 = <u>1440.</u>

For the number of games that is over in 6 moves, 3 "X" and 2 "O" on the game. Thus, the possibilities for the number of games that is over in 6 moves is 8 * 3 * 2 * 6 * 5 * 4 = 5760. However, it includes the cases which 3 "X" and 3 "O" are in a row. We only care for 3 horizontals and 3 verticals since there is no possibilities of 3 "X" are in a row when 3 "O" are diagonally placed. So, it's 6 * 3 * 2 for 3 horizontal and 3 vertical rows for 3 "O" and 2 * 3 * 2 for 3 "X", which is 432. Thus, the number of games that is over in 6 moves are <u>5328.</u>

For the number of games that is over in 7 moves, 4 "X" and 3 "O" are on the game. For 4 "X", it's 8 * 3 * 6 * 3 * 2 and For 3 "O", it's 5 * 4 * 3. Thus, the total is 8 * 3 * 6 * 3 * 2 * 5 * 4 * 3 = 51840. But like the number of games that is over in 6 moves, We need to deduct some cases that is 3 "X" and 3 "O" are in a row. And it's 6 * 3 * 6 * 3* 2 * 3 * 2 = 3888. Thus, the number of games that is over in 7 moves are 51840 – 3888 = <u>47952</u>

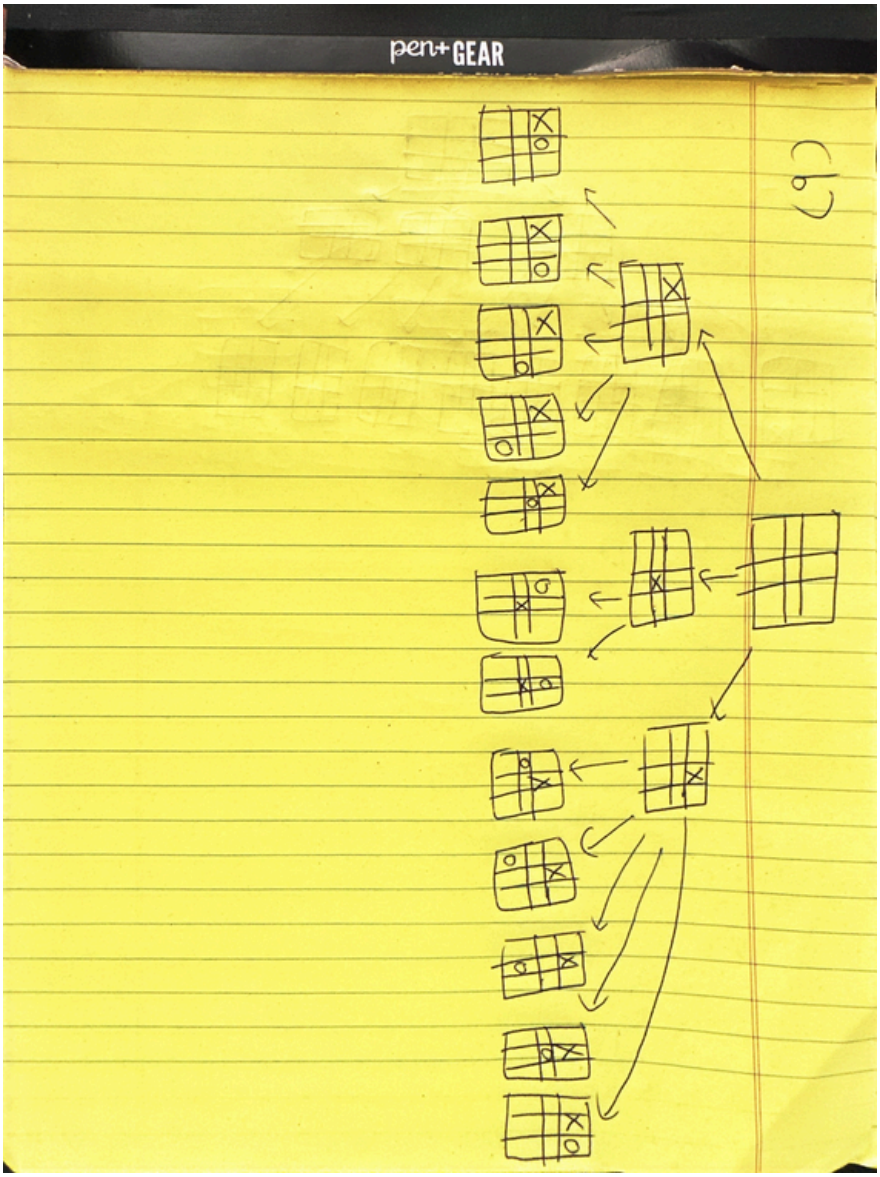
For the number of games that is over in 8 moves, it's 8 * 3 * 6 * 3 * 2 * 5 * 4 * 3 * 2 = 103680 and deduct same cases above, 6 * 3 * 6 * 3 * 2 * 2 * 4 * 3 * 2 = 31104. So, it's 103680 – 31104 = <u>72576.</u>

For the number of games that is over in 9 moves, there are two options which is draw & win

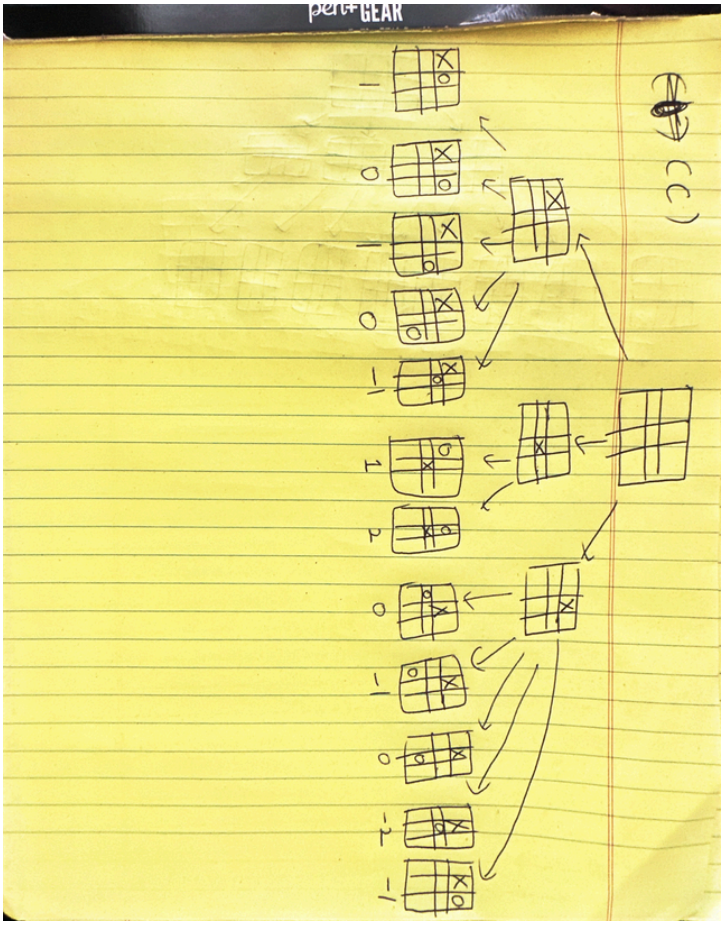
For x to win vertically or horizontally, it's 6 * 3 * 4 * 4 * 3 * 2 * 4 * 3 * 2 = 41472. For X to win diagonally, it's 2 * 3 * 8 * 4 * 3 * 2 * 4 * 3 * 2 = <u>27648</u>. For x to win by becoming two distinct 3 "X" in a row, it's 22 * 4 * 3 * 2 * 4 * 3 * 2 = <u>12672</u>. And draw -> it's 16 * 5 * 4 * 3 * 2 * 4 * 3 * 2 = <u>46080</u>.

<u>Thus, possible games of tic tac toe are 255168.</u>

(b)

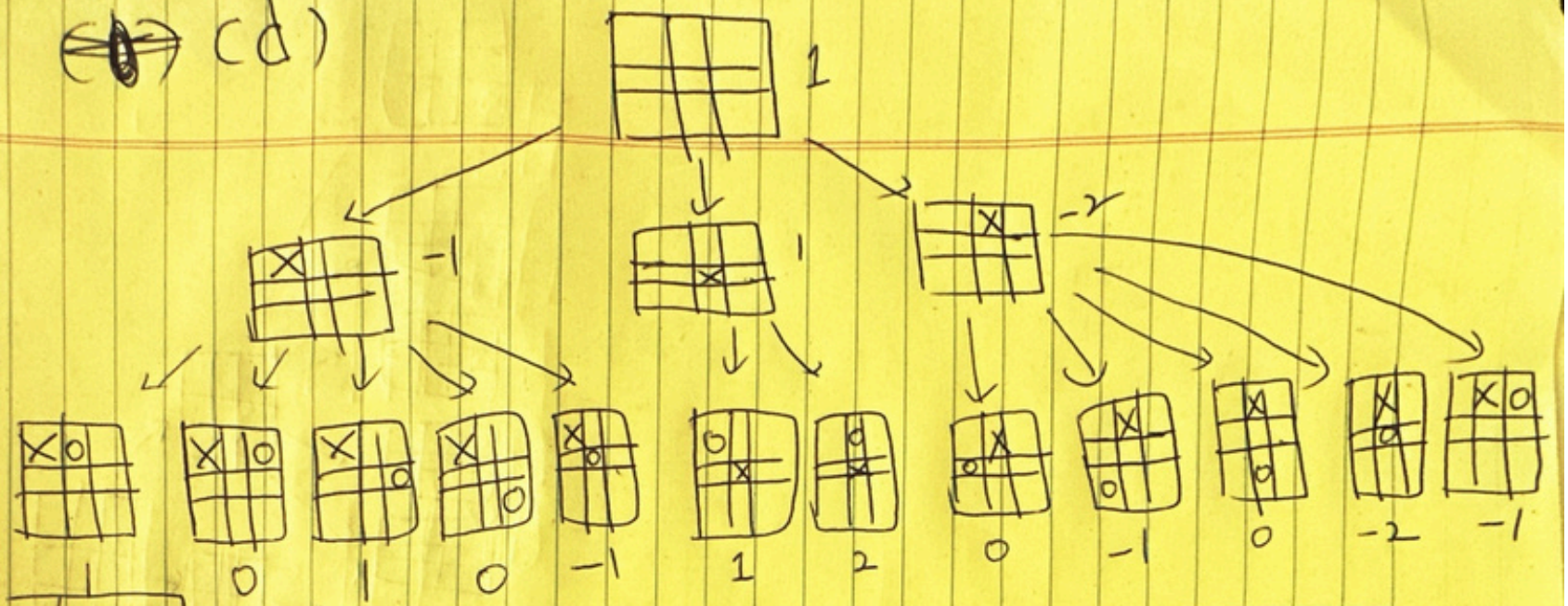


(c)

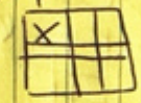


(d)

(d)



depth = 1 → min nodes

For  , the back-up value is $\min\{1, 0, 1, 0, -1\} = -1$

For , the back-up value is $\min\{1, 2\} = 1$

For , the back-up value is $\min\{0, -1, 0, -2, -1\} = -2$

depth = 0 → max nodes

↓

Backup value → $\max(-1, 1, 2)$

⇓

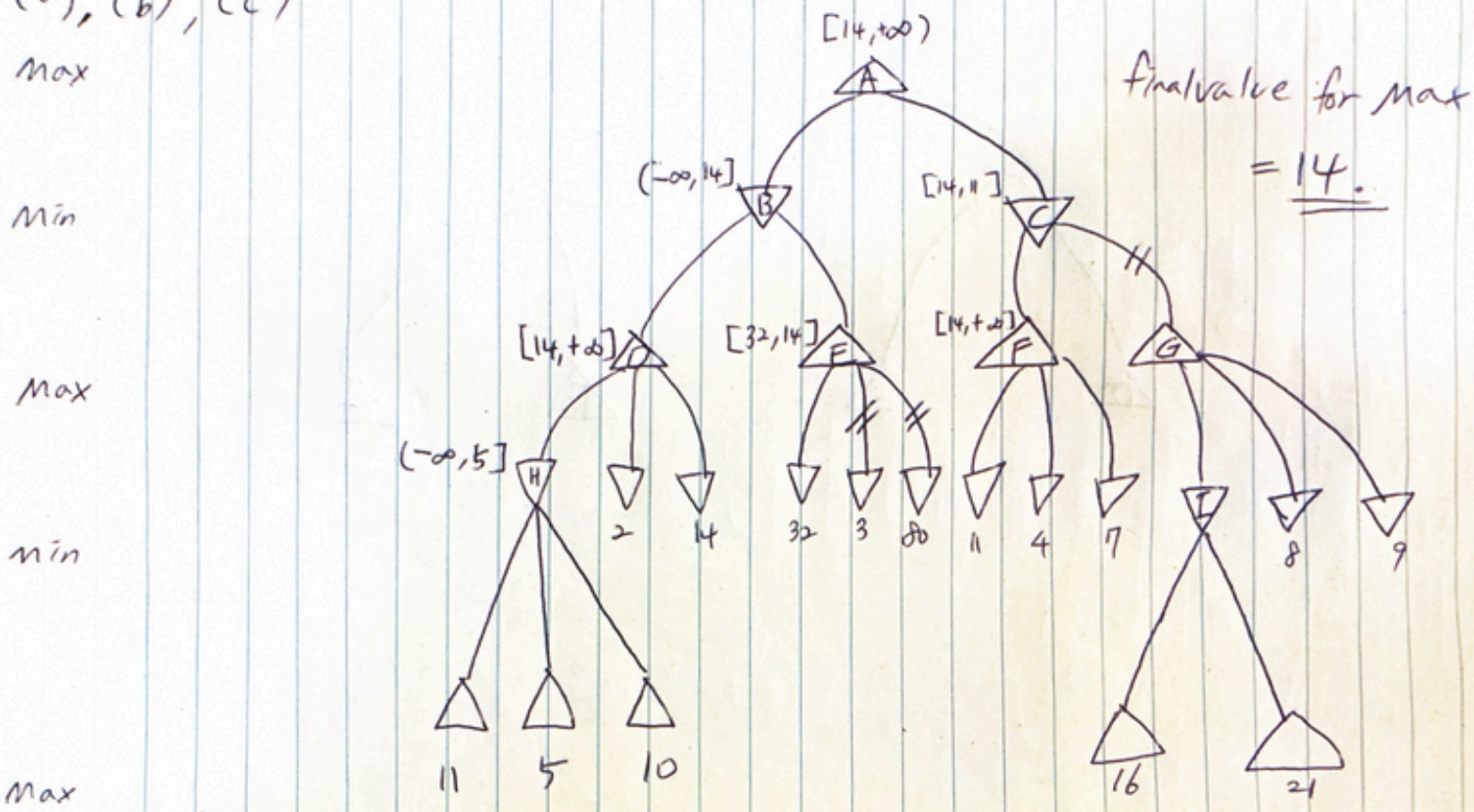
1.

⇓

Thus, is the best starting move.

2.

(a), (b), (c)



finalvalue for max

$= \underline{\underline{14}}.$

variables: X = {WA, NT, Q NSW, V, SA, T}

Let Domains: D {red(R), green(G), blue(B), orange(O)} (4 colors)

1. Start from WA => R,G,B, or O => 4 choices
2. NT => 3 choices
3. SA => 2 choices
4. Q => 2 choices
5. NSW => 2 choices
6. V => 2 choices
7. T => 4 choices

Thus, there are 4 * 3 * 2 * 2 * 2 * 2 * 4 solutions for four colors.

<u>768 solutions if 4 colors are allowed and there are 0 solution if 2 colors are allowed.</u>